EXP-12 : To encrypt and decrypt the given message using the 2 x 2 Hill key.

PROGRAM :

Run | Debug | Stop | Share | Save | { } Beautify

main.c

```c
#include <stdio.h>
#include <string.h>
void hill(char *s, char *out, int k[2][2]) {
int i;
for (i = 0; s[i]; i += 2) {
int x = s[i]-'A', y = s[i+1]-'A';
out[i] = (k[0][0]*x+k[0][1]*y)%26+'A';
out[i+1] = (k[1][0]*x+k[1][1]*y)%26+'A';
}
out[i] = 0;
}
int main(void) {
char p[100] = "MEETMEATTHEUSUALPLACEATTENRATHER"
"THANEIGHTOCLOCK";
char c[100], d[100];
int k[2][2] = {{9,4},{5,7}};
int inv[2][2] = {{5,12},{15,25}};
int n = (int)strlen(p);
if (n % 2) { p[n] = 'X'; p[n+1] = 0; }
hill(p,c,k); hill(c,d,inv);
d[n] = 0; /* remove only the padding added above */
printf("Encrypted: %s\nDecrypted: %s\n", c,d);
return 0;
}

```

OUTPUT :

input

```
Encrypted: UKIXUKYDROMEIWSZXWIOKUNUKHXHROAJROANQYEBTLKJEGAD
Decrypted: MEETMEATTHEUSUALPLACEATTENRATHERTHANEIGHTOCLOCK


...Program finished with exit code 0
Press ENTER to exit console.
```